Chapter 14. Subsequence Conditions on Permutations

So Hard to Avoid

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14.1 Pattern avoidance

Motivation. Children of distinct heights line up so that each child sees every shorter child ahead of them: the lineup must contain no entries $a \dots b \dots c$ with $a < c < b$. Counting permutations that avoid such a "pattern" produces the Catalan numbers for every pattern of length 3. For length 4 the patterns behave differently, and only exponential bounds are within easy reach.

In 1423567, child 2 cannot see 1 (blocked by 4); 6723415 is a good lineup. A lineup is good iff there are no entries a, b, c, in this order (not necessarily adjacent), with $a < c < b$.

Definition 14.1 (132-pattern)

Entries a, b, c of a permutation, in this left-to-right order, with $a < c < b$ form a 132**-pattern** (they are ordered like $1, 3, 2$). Similarly for 123, 231, etc.

Definition 14.2 (132-avoiding)

A permutation with no 132-pattern is 132**-avoiding**.

Let $f(n)$ count 132-avoiding n-permutations ($f(0) = 1$). If n is in position i, every entry left of n exceeds every entry right of n (else $x < y$ with x left and y right gives the pattern $x\,n\,y$). So the right side is a 132-avoiding permutation of $\{1, \dots, n-i\}$ and the left side one of $\{n-i+1, \dots, n-1\}$, and conversely any such pair gives a 132-avoiding permutation. Hence

$$f(n) = \sum_{i=1}^{n} f(i-1) f(n-i) \qquad (n \geq 1). \tag{14.1}$$

Corollary 14.3

The number of 132-avoiding n-permutations is the Catalan number $c_n = \frac{1}{n+1}\binom{2n}{n}$.

Proof. (14.1) with $f(0) = 1$ is the recurrence of the Catalan numbers: comparing coefficients in (8.6) gives $c_n = \sum_{i=1}^{n} c_{i-1} c_{n-i}$ (§8.1.2.1). The two sequences agree by induction, and $c_n = \frac{1}{n+1}\binom{2n}{n}$. ■

Definition 14.4 (Pattern)

Let p be an n-permutation and $q = q_1 \cdots q_k$ a k-permutation, $k \leq n$. Entries $a_1, \dots, a_k$ of p (left to right) form a q**-pattern** if $a_i < a_j \iff q_i < q_j$.

Definition 14.5 (Avoidance)

p is q**-avoiding** if no k entries of p form a q-pattern. $S_n(q)$ (or $\mathrm{Av}_n(q)$) denotes the number of q-avoiding n-permutations.

Symmetries. The **reverse** $p_n \cdots p_1$ and the **complement** $\bar{p} = (n+1-p_1) \cdots (n+1-p_n)$ map q-avoiders bijectively onto avoiders of the reversed or complemented pattern. 231 is the reverse of 132, 312 its complement, and 213 the reverse of 312. Hence

$$S_n(132) = S_n(231) = S_n(312) = S_n(213), \qquad S_n(123) = S_n(321).$$

An entry smaller than all entries before it is a **left-to-right minimum**; these entries form a decreasing subsequence that starts with p_1 and contains 1.

Lemma 14.6

$S_n(123) = S_n(132)$ for all $n \geq 1$.

Proof. We construct a bijection from 123-avoiders to 132-avoiders that fixes the left-to-right minima (values and positions).

Step 1 (constraints). Let $m_1 > m_2 > \cdots > m_r$ be left-to-right minima at positions $i_1 = 1 < i_2 < \cdots < i_r$; call the other positions *slots*, a slot between i_j and i_{j+1} having *threshold* m_j. Filling the slots with the remaining values gives a permutation with exactly these left-to-right minima iff every slot receives a value larger than its threshold. Thresholds decrease from left to right.

Step 2 (the map f). Given a 123-avoider p, fill the slots from left to right, each time with the smallest unused value exceeding the threshold. This never gets stuck: at the s-th slot with threshold m, all of the first s slots have thresholds $\geq m$, so in p they hold s values $> m$. Only $s-1$ values have been used, so a value $> m$ remains. By Step 1, $f(p)$ has the same left-to-right minima.

Step 3 ($f(p)$ avoids 132). If non-minima x before y both exceed a left-to-right minimum m_j to their left, then when x was placed (threshold $\leq m_j < y$), y was available, so $x < y$. Now let a, b, c form a 132-pattern. Replacing a by the last left-to-right minimum at or before it keeps the pattern, so let $a = m_j$. Then $b, c > m_j$ lie right of m_j, so they are not left-to-right minima, and by the observation $b < c$, a contradiction.

Step 4 (the inverse g). Given a 132-avoider q, fill its slots with the non-minima in *decreasing* order. This respects the thresholds: in q the first s slots hold s values exceeding the s-th threshold, so the s-th largest value does too. $g(q)$ is a union of two decreasing sequences, hence avoids 123.

Step 5 (g and f are inverse). A 123-avoider has its non-minima in decreasing order (if $x < y$ were non-minima in this order, the left-to-right minimum before x, smaller than x, together with x, y would form a 123), so $g(f(p)) = p$. A 132-avoider q equals the greedy filling of Step 2. Indeed, if at some slot (threshold m_j) q holds x while a smaller eligible unused $y > m_j$ comes later, then m_j, x, y is a 132-pattern. Hence $f(g(q)) = q$. ■

For example, $p = 465132$ has minima $4, 1$ and $f(p) = 456123$.

Theorem 14.7

For every pattern q of length 3 and all $n \geq 1$, $S_n(q) = c_n = \frac{1}{n+1}\binom{2n}{n}$.

Proof. By the symmetries and Lemma 14.6, all six values $S_n(q)$ coincide, and $S_n(132) = c_n$ by Corollary 14.3. ■

For length 4, symmetries and further arguments leave three classes: 1342, 1234, 1324, with

$$\begin{aligned} S_n(1342): &\quad 1, 2, 6, 23, 103, 512, 2740, 15485 \\ S_n(1234): &\quad 1, 2, 6, 23, 103, 513, 2761, 15767 \\ S_n(1324): &\quad 1, 2, 6, 23, 103, 513, 2762, 15793 \end{aligned}$$

for $n \leq 8$. In fact $S_n(1342) < S_n(1234) < S_n(1324)$ for $n \geq 7$; surprisingly, the monotone pattern is in the middle. We prove the second inequality.

Theorem 14.8

$S_n(1234) < S_n(1324)$ for all $n \geq 7$.

A **right-to-left maximum** is an entry larger than all entries to its right. The proof is given after Lemma 14.11.

Definition 14.9 (Class)

Permutations x, y are in the same **class** if they have the same left-to-right minima at the same positions and the same right-to-left maxima at the same positions.

For instance 51234 and 51324 are in the same class; 24315 and 24135 are not.

Constraints of a class [Supplement]. Fix a class and call the remaining positions slots. A slot s has a lower bound $l(s)$, the nearest left-to-right minimum to its left, and an upper bound $u(s)$, the nearest right-to-left maximum to its right. A filling of the slots lies in the class iff $l(s) <$ value $< u(s)$ for every slot. (Then no slot entry becomes a new minimum or maximum, and the given minima and maxima keep their status.) Both l and u are non-increasing from left to right.

Lemma 14.10

Every nonempty class contains exactly one 1234-avoiding permutation.

Proof.

Step 1 (existence). Fill the slots with the remaining values in decreasing order. This respects the bounds [Supplement]. Take any member of the class. Its first s slots hold s values exceeding $l(s)$, so the s-th largest value exceeds $l(s)$. Its last $N - s + 1$ slots (of N) hold values below $u(s)$, so the $(N - s + 1)$-th smallest, which is the s-th largest, is below $u(s)$. The result is a union of three decreasing sequences (minima, maxima, slots), so a 1234-pattern would have two entries in one of them, impossible by pigeon-hole.

Step 2 (uniqueness). In a 1234-avoider of the class, if slot entries $a < b$ appeared in this order, then the left-to-right minimum before a (smaller than a), a, b, and the right-to-left maximum after b (larger than b) would form 1234. So the slot entries are decreasing, and the filling is the one of Step 1. ∎

An **inversion** of p is a pair $i < j$ with $p_i > p_j$.

Lemma 14.11

Every nonempty class contains at least one 1324-avoiding permutation.

Proof.

Step 1 (good patterns). If p contains a 1324-pattern $a\,c\,b\,d$ ($a < b < c < d$), replace a by the nearest left-to-right minimum at or before it and d by the nearest right-to-left maximum at or after it. The result is still a 1324-pattern, now starting at a left-to-right minimum and ending at a right-to-left maximum (a **good pattern**). The middle entries c, b are slot entries: a left-to-right minimum can only play the role of "1" and a right-to-left maximum only that of "4".

Step 2 (swapping). Exchange c and b. The value b moves left, with $b > a \geq$ the new lower bound and $b < c < u$ there. The value c moves right, with $c < d \leq$ the new upper bound and $c > b > l$ there. So the bounds hold, and the new permutation is in the same class. Swapping an inverted pair reduces the number of inversions by at least one.

Step 3 (termination). Repeat while a good pattern exists. After at most $\binom{n}{2}$ steps, we reach a permutation of the same class with no good pattern, hence (Step 1) no 1324-pattern. ∎

Proof of Theorem 14.8. By Lemmas 14.10 and 14.11, $S_n(1234) \leq S_n(1324)$; it remains to find a class with two 1324-avoiders. Write $a_1{*}a_2{**}b_1$ for the class of length-6 permutations with left-to-right minima a_1, a_2 at positions $1, 3$ and right-to-left maximum b_1 at position 6. For $n = 7$ the class $3{*}1{*}7{*}5$ contains the two 1324-avoiders 3612745 and 3416725. For $n > 7$ use the class $n(n-1)\cdots 8\,3{*}1{*}7{*}5$. ∎

Stanley–Wilf. For every pattern q there is c_q with $S_n(q) \le c_q^n$ (Marcus–Tardos, 2004; cited). For monotone patterns this is easy:

Theorem 14.12

For $k \le n$, $S_n(12\cdots k) \le (k-1)^{2n}$.

Proof. Say x has **rank** i if the longest increasing subsequence ending at x has length i. Entries of equal rank form a decreasing subsequence (if $x < y$ had equal rank with x first, appending y would raise its rank). A $12\cdots k$-avoider has ranks in $[k-1]$, so it is a union of $k-1$ decreasing subsequences. The permutation is determined by assigning each value to one of the $k-1$ subsequences ($(k-1)^n$ ways) and each position to one of them ($(k-1)^n$ ways): each subsequence's values then fill its positions in decreasing order. ∎

This agrees with $S_n(123) = c_n < 4^n$.

Definition 14.13 (Direct sum)

For $p \in S_a$, $q \in S_b$, the **direct sum** $p \oplus q \in S_{a+b}$ is p followed by q with every entry increased by a.

Example 14.14

$132 \oplus 2431 = 1325764$.

Theorem 14.15

If $S_n(q_1 \oplus 1) \le c_1^n$ and $S_n(1 \oplus q_2) \le c_2^n$ for all n, then

$$S_n(q_1 \oplus 1 \oplus q_2) \le \left(\sqrt{c_1} + \sqrt{c_2}\right)^{2n}.$$

Example 14.16

$S_n(2134) = S_n(1234) < 9^n$ (the equality is a known Wilf-equivalence, cited; the bound is Theorem 14.12), and $S_n(1243) < 9^n$ likewise. With $q_1 = 213$, $q_2 = 132$, Theorem 14.15 gives $S_n(213 \oplus 1 \oplus 132) \le (3+3)^{2n} = 36^n$.

Proof of Theorem 14.15. Let p avoid $q = q_1 \oplus 1 \oplus q_2$. Colour an entry red if it is the last (largest) entry of some $(q_1 \oplus 1)$-pattern, blue otherwise.

Step 1. The red entries avoid $1 \oplus q_2$. Given a red copy C, its first entry x ends a copy of $q_1 \oplus 1$ whose other entries lie left of x and are smaller than x, hence smaller than all of C. Together these give a copy of q.

Step 2. The blue entries avoid $q_1 \oplus 1$: the last entry of such a copy would be red.

Step 3 (counting). With k blue entries, p is determined by the set of blue values, the set of blue positions, the pattern of the blue entries and that of the red ones: at most $\binom{n}{k}^2 c_1^k c_2^{n-k}$ choices. Since a sum of squares of nonnegative numbers is at most the square of their sum,

$$S_n(q) \le \sum_{k=0}^{n} \left(\binom{n}{k} \sqrt{c_1}^{\,k} \sqrt{c_2}^{\,n-k}\right)^2 \le \left(\sum_{k=0}^{n} \binom{n}{k} \sqrt{c_1}^{\,k} \sqrt{c_2}^{\,n-k}\right)^2 = \left(\sqrt{c_1} + \sqrt{c_2}\right)^{2n}.$$

∎

14.2 Stack sortable permutations

Motivation. Passing a permutation once through a stack sorts it exactly when it avoids 231. Sorting twice is no longer described by plain pattern avoidance, but the recursion $s(LnR) = s(L)s(R)n$, read on decreasing binary trees, reveals a hidden symmetry: t-stack sortable permutations with k descents are as numerous as those with $n-1-k$ descents.

Stack sorting. Read p from left to right. An incoming entry is pushed on the stack if it is smaller than the top; otherwise the top is popped to the output, repeatedly, until the entry can be pushed. At the end the stack is emptied. The output is $s(p)$; p is **stack sortable** if $s(p) = 12\cdots n$.

Example 14.17

$p = 2413$: push 2; 4 pops 2 and is pushed; push 1; 3 pops 1 and is pushed on 4; finally pop $3, 4$. So $s(2413) = 2134$, and 2413 is not stack sortable.

Proposition 14.18

Let $a < b$ be entries of p.
(1) If a precedes b in p, then a precedes b in $s(p)$.
(2) If b precedes a and no entry $c > b$ lies between them, then a precedes b in $s(p)$.
(3) If b precedes a and some $c > b$ lies between them (so b, c, a form a 231-pattern), then b precedes a in $s(p)$.

Proof.

(1) b cannot be pushed on top of the smaller a, so a leaves the stack before b enters.

(2) An entry leaves the stack only when a larger entry arrives or at the end. All entries between b and a are smaller than b, so b is still in the stack when a is pushed above it, and a leaves first. [Supplement] (The book's claim that the entries between are decreasing is not needed.)

(3) b must leave before c is pushed, and c is pushed before a arrives. ■

Theorem 14.19

p is stack sortable iff it avoids 231.

Proof. A 231-pattern bca puts b before a in $s(p)$ by part (3). Without such patterns every pair $a < b$ falls under (1) or (2) and appears in increasing order in $s(p)$. ■

So the stack sortable n-permutations are counted by c_n (Theorem 14.7). p is **two-stack sortable** if $s(s(p))$ is the identity. This property is not closed under taking patterns:

Example 14.20

$s(35241) = 32145$ and $s(32145) = 12345$, so 35241 is two-stack sortable. But its subsequence 3241 has $s(3241) = 2314$ and $s(2314) = 2134$.

Theorem 14.21 (West)

p is two-stack sortable iff it contains no 2341-pattern and no 3241-pattern except as part of a 35241-pattern.

Proof.

Step 1 (a 2341 obstructs). Let $b\,c\,d\,a$ form 2341 ($a<b<c<d$). By Proposition 14.18, b precedes c (part 1), and b and c precede a in $s(p)$ (part 3, via d). So b, c, a is a 231 in $s(p)$, and $s(p)$ is not stack sortable.

Step 2 (a bad 3241 obstructs). Let $y\,x\,z\,w$ form 3241 ($w<x<y<z$), not inside a 35241. Via z, both x and y precede w in $s(p)$. If no entry between y and x exceeds y, then x precedes y in $s(p)$ (part 2), and x, y, w is a 231 in $s(p)$. If some t between y and x exceeds y, then $t<z$ (otherwise $y\,t\,x\,z\,w$ is a 35241), so $y\,t\,z\,w$ is a 2341, and Step 1 applies.

Step 3 (sufficiency). Suppose $s(p)$ contains a 231-pattern $f\,g\,e$ ($e<f<g$). Since e comes after f and g in $s(p)$, Proposition 14.18 shows that f, g precede e in p and that some $h>g$ lies between g and e in p.

(a) If f precedes g in p, then $f\,g\,h\,e$ is a 2341.

(b) If g precedes f: as f precedes g in $s(p)$, no entry between g and f exceeds g (part 3). Hence h lies between f and e, and $g\,f\,h\,e$ is a 3241. It is not part of a 35241, which would need an entry larger than $h>g$ between g and f. ■

The number of two-stack sortable n-permutations is $\frac{2}{(n+1)(2n+1)}\binom{3n}{n}$ (cited). p is **t-stack sortable** if $s^t(p)$ is the identity.

Lemma 14.22

If $p = LnR$, then $s(p) = s(L)\,s(R)\,n$.

Proof. The entries of L all leave the stack before n enters, producing $s(L)$. Then n sits at the bottom while R passes over it, producing $s(R)$, and n leaves last. ■

Corollary 14.23

Every n-permutation is $(n-1)$-stack sortable.

Proof. Induction on n. By Lemma 14.22, $s(p) = s(L)s(R)\,n$ and $s(\sigma n) = s(\sigma)\,n$ for any σ. Hence $s^{n-1}(p) = s^{n-2}(s(L)s(R))\,n$, and $s(L)s(R)$ is a permutation of $[n-1]$, which is $(n-2)$-stack sortable by induction. ■

Corollary 14.24

$s^t(p)$ ends with $(n-t+1)(n-t+2)\cdots n$.

Proof. Induction on t [Supplement]. If $s^{t-1}(p) = \sigma\,(n-t+2)\cdots n$ with σ a permutation of $[n-t+1]$, then by Lemma 14.22 (applied to the trailing maxima one by one), $s^t(p) = s(\sigma)\,(n-t+2)\cdots n$, and $s(\sigma)$ ends with $\max\sigma = n-t+1$. ■

Decreasing binary trees. $T(p)$ has root n. The maximum of L is its left child and the maximum of R its right child (a missing side gives no child, but an only child is still a left or a right child). Recursively, $T(L)$ and $T(R)$ hang below. Reading $T(p)$ *in order* (left subtree, root, right subtree) recovers p.

Example 14.25

For $p = 263498175$, $T(p)$ is drawn below.

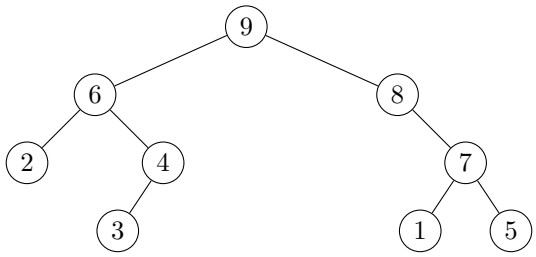


Figure. $T(263498175)$. *In order:* 263498175*; postorder:* $234615789 = s(p)$. *Here* 4 *has only a left child and* 8 *only a right child.*

Example 14.26

Reading the tree above in *postorder* (left subtree, right subtree, root) gives 234615789, which is $s(263498175)$.

Proposition 14.27

$T(p)$ read in postorder is $s(p)$.

Proof. Induction on n. For $p = LnR$, postorder reads $T(L)$, then $T(R)$, then n, giving $s(L)s(R)n$ by induction, which is $s(p)$ by Lemma 14.22. ■

i is a **descent** of p if $p_i > p_{i+1}$; $d(p)$ is the number of descents. Complementation shows that k and $n-1-k$ descents are equally frequent among all n-permutations. [Supplement] $p_i > p_{i+1}$ iff p_i has a right child in $T(p)$: the in-order successor of a vertex with a right child is the leftmost vertex of its right subtree (smaller), and otherwise it is an ancestor (larger). So $d(p)$ equals the number of right edges of $T(p)$.

Let $W_t(n,k)$ be the number of t-stack sortable n-permutations with k descents. Neither reverse nor complement preserves t-stack sortability (try 213 and 132), yet the data suggest $W_t(n,k) = W_t(n,n-1-k)$.

Definition 14.28 (The map f)

Define f on permutations (of any finite set of integers) by $f(1) = 1$ and, with n the maximum: $f(LnR) = f(L)\,n\,f(R)$ if L, R are nonempty; $f(Ln) = n\,f(L)$; $f(nR) = f(R)\,n$.

Example 14.29

$f(123) = 3\,f(12) = 3\,2\,f(1) = 321$, so $f(4123) = f(123)\,4 = 3214$.

Example 14.30

$f(1423) = f(1)\,4\,f(23) = 1432$.

Example 14.31

$f(412395867) = f(4123)\,9\,f(5867) = 3214\,9\,5876$.

Proposition 14.32

$d(p) + d(f(p)) = n - 1$ for every n-permutation p.

Proof. Induction on n; $n = 1$ is trivial.

Step 1 ($p = LnR$, $L, R \neq \emptyset$, n in position i). $d(p) = d(L) + d(R) + 1$ and $d(f(p)) = d(f(L)) + d(f(R)) + 1$, while by induction $d(L) + d(f(L)) = i - 2$ and $d(R) + d(f(R)) = n - i - 1$. The total is $n - 1$.

Step 2 ($p = Ln$). $d(p) = d(L)$ and $d(n\,f(L)) = d(f(L)) + 1$, total $(n-2)+1$.

Step 3 ($p = nR$). $d(p) = 1 + d(R)$ and $d(f(R)\,n) = d(f(R))$, total $1 + (n-2)$. ■

Lemma 14.33

$s(p) = s(f(p))$ for every permutation p.

Proof. Induction on n, using Lemma 14.22 and the induction hypothesis $s(L) = s(f(L))$, $s(R) = s(f(R))$ (f permutes the entries of L and R among themselves).

(1) $p = LnR$: $s(p) = s(L)s(R)n = s(f(L))s(f(R))n = s(f(L)nf(R)) = s(f(p))$.

(2) $p = nR$: $s(p) = s(R)n = s(f(R))n = s(f(R)n) = s(f(p))$.

(3) $p = Ln$: $s(p) = s(L)n = s(f(L))n = s(nf(L)) = s(f(p))$. ■

Corollary 14.34

p is t-stack sortable iff $f(p)$ is.

Proof. Both are equivalent to $s(p) = s(f(p))$ being $(t-1)$-stack sortable. ■

Theorem 14.35

$W_t(n,k) = W_t(n, n-1-k)$ for all n, k, t.

Proof. f is a bijection (it acts on $T(p)$ by flipping every only child to the other side, which is an involution on trees; see below). By Corollary 14.34 and Proposition 14.32 it maps t-stack sortable permutations with k descents onto those with $n-1-k$ descents. ■

Tree view. On $T(p)$, f turns every only left child into an only right child and vice versa, and keeps vertices with two children unchanged. This swaps the numbers of left and right edges, which is Proposition 14.32 again. For postorder it does not matter on which side an only child hangs, which is Lemma 14.33 again, and in fact holds for the subtree of every vertex.

Example 14.36

$p = 356124$ and $f(p) = 536421$ both give 351246 when their trees are read in postorder, and the same holds for the subtree of each entry.